FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2015

ANSWERS, HINTS & SOLUTIONS
FULL TEST–II
(PAPER-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	C	B
2.	B	A	C
3.	A	D	D
4.	B	A	B
5.	A	C	D
6.	C	D	D
7.	D	C	C
8.	A, B	A, B, C, D	C, D
9.	A, B, C	A, B, C	B, C
10.	A, B	B, D	A, B, D
11.	A, B, D	C, D	A, C
12.	*A*	B	B
13.	A	B	D
14.	A	B	B
15.	A	C	A
16.	B	D	C
1.	(A) → (q), (B) → (p, q, s, t), (C) → (p, q, s, t), (D) → (r, s)	(A) – (s), (B) – (r), (C) – (q), (D) – (p)	(A) → (p, q, s) (B) → (t) (C) → (s) (D) → (q)
2.	(A) →(r), (B) → (r), (C) → (p), (D) → (p)	(A – r), (B – s), (C – p,), (D – q)	(A) → (p) (B) → (q) (C) → (p) (D) → (t)
1.	3	2	3
2.	2	5	4
3.	7	2	2
4.	2	6	7
5.	5	2	8
6.	4	4	5

Physics

PART – I

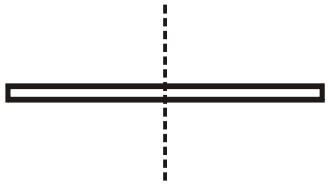
SECTION – A

1. Momenta and charged are equal and opposite, so sense of the motions are opposite.

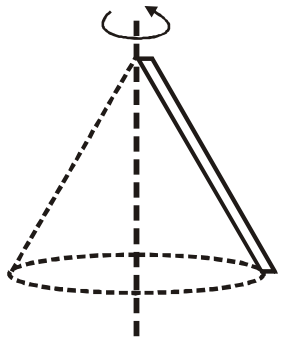
$\therefore R = \frac{Mv}{qB} \Rightarrow$ their radii are equal.

2. We can not assume total mass to be concentrated at centre of mass for calculation of all the thing e.g. for calculation of moment of inertia, calculation of torque of centrifugal force on a rod about hinge as shown.

(i)



(ii)



6. $Kx = (m + M)\, a + ma' \cos\theta$

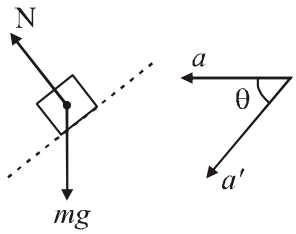
Block: $a' + a\cos\theta = g\sin\theta$ ...(ii)

From (i) & (ii)

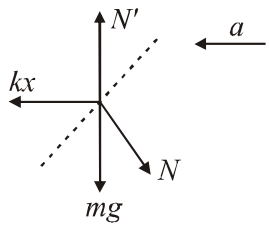
$$a = \frac{Kx - mg\sin\theta\cos\theta}{M + m\sin^2\theta} = K_1 x - K_2$$

$\Rightarrow$ Max. compression $= \frac{2mg\sin\theta\cos\theta}{K}$

Block :



Wedge :



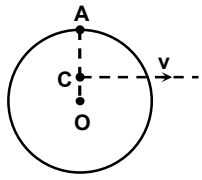
Block + Wedge

7. Net magnetic moment of loop formed is zero, hence torque due to magnetic field is zero.

11. $K = K_D + K_P$

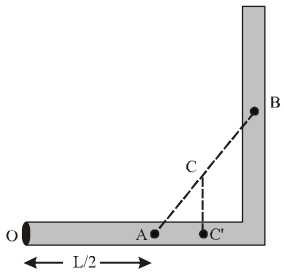
$$= \frac{1}{2}mv_0^2\left(1+\frac{1}{2}\right) + \frac{1}{2}m(2V_0)^2$$

$$= \frac{11}{4}mV_0^2$$



SECTION – C

3. Let A, B, and C be C.M. of larger section , smaller section and total rod



$$CC' = \frac{L}{12}, AC' = \frac{L}{6}$$

$OC = \left\{\left(\frac{L}{2}+\frac{L}{6}\right)^2+\left(\frac{L}{12}\right)^2\right\}^{\frac{1}{2}}$ =

$\frac{\sqrt{65}}{12}L$

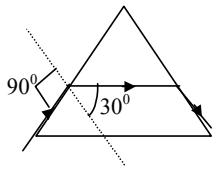
Moment of inertia about '*O*'

$I = \frac{7}{8}ML^2 \Rightarrow$ Maximum torque of about '*O*'

$$\tau_{max} = \frac{3Mg}{2}\times(OC)$$

$$\alpha_{max} = \frac{\tau_{max}}{I} = \frac{\sqrt{65}}{7}\left(\frac{g}{L}\right)$$

4. $\mu = \frac{\sin 90^0}{\sin 30^0} = 2$



5. $kx_0 = ma_0$

6. $\alpha = \tau/I = \left(Mg\frac{L}{2}\Big/\frac{ML^2}{3}\right) = \frac{3}{2}\frac{g}{L}$

$\Rightarrow a_{cm} = \alpha L/2 = 3g/4 = \frac{Mg-N}{M}$

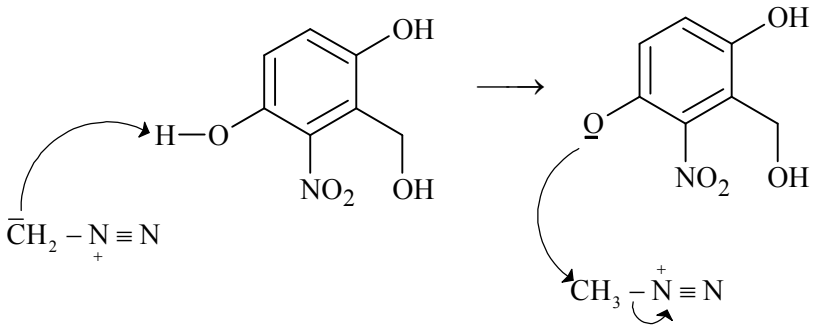
$\Rightarrow N = \frac{Mg}{4}$

Chemistry

PART – II

SECTION – A

1. Diazomethane being a base will pick up the hydrogen which is most acidic. The H of the OH group ortho to $-NO_2$ group is most acidic



2. Choice (B) is incorrect as all the gases above their respective Boyle's temperature show throughout +ve deviation.

4. All reducing sugars are mutarotating. Although IV is an α-hydroxy ketone and hence reducing but it can't mutarorate as it is not a carbohydrate/can't form ring. In II the glycosidic bond is in between two anomeric carbons and hence ring opening can't occur. Thus non-reducing as well as non-mutarotating.

5. $$\underset{(A)}{Mg_3N_2} + 6H_2O \longrightarrow 3Mg(OH)_2 + \underset{}{2NH_3}$$

$$2NH_3 + 3CuO \longrightarrow 3Cu + \underset{(C)}{N_2} + 3H_2O$$

$$NaNO_2 + Zn + NaOH \longrightarrow Na_2ZnO_2 + NH_3 + H_2O$$

Ammonium salts containing anions which are oxidizing in nature, on heating liberates $N_{2(g)}$ instead of NH_3.

9. At constant pressure, the addition of inert gas shifts the equilibrium towards larger no. of moles.

14. Since, it liberates a gas on reaction with dilute H_2SO_4, it cannot be a sulphate salt, it would be a carbonate salt. Malachite ore of copper, $CuCO_3$. $Cu(OH)_2$ is green in colour.

15. $$\underset{\text{Blue colour}}{\left[Cu(NH_3)_4\right]^{2+}} + 2KCN \longrightarrow \underset{\text{Yellow ppt}}{Cu(CN)_2\downarrow} + 2K^+ + 4NH_3$$

$$2Cu(CN)_2\downarrow \xrightarrow{\text{decomposes}} \underset{\text{White ppt}}{2CuCN\downarrow} + (CN)_2\uparrow$$

$$CuCN\downarrow + 3KCN \longrightarrow \underset{\text{Colourless}}{\left[Cu(CN)_4\right]^{3-}} + 3K^+$$

Cu^{+1} has a d^{10} configuration, so it is tetrahedral in shape with sp^3 hybridization.

16. Gravimetric estimation of Cu^{2+} would be possible only when the given procedure gives an insoluble compound on treatment with a given reagent.

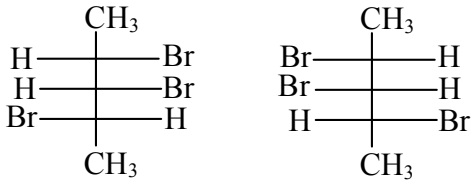
$$Cu^{2+} + 2KSCN \longrightarrow \underset{\text{Black ppt}}{Cu(SCN)_2 \downarrow} + 2K^+$$

$$2Cu(SCN)_2 \downarrow \xrightarrow{\text{decomposes}} \underset{\text{White ppt}}{2CuSCN \downarrow} + (SCN)_2 \uparrow$$

With other reagents, either there is no reaction or even if it reacts, then the product is a soluble compound.

SECTION –C

1. Only two optically active isomers are possible.



2. $\dfrac{r_{C_nH_{2n+2}}}{r_{H_2O}} = \dfrac{1}{2} = \sqrt{\dfrac{M_{H_2O}}{M_{C_nH_{2n+2}}}}$

$\therefore\ M_{C_nH_{2n+2}} = 4 \times 18 = 72$

$\therefore\ 12 \times n + 1 \times (2n + 2) = 72$

$\therefore\ 14n + 2 = 72$

$\therefore\ n = 5$

3. Let formula of hydrated dibasic acid be $H_2A\ .x\ H_2O$

Equivalent of dibasic acid = Equivalent of NaOH

$$\frac{2.505 \times 2}{M} = \frac{35}{1000} \times 1$$

$$M = \frac{2.505 \times 2 \times 1000}{35 \times 1} = 143.14$$

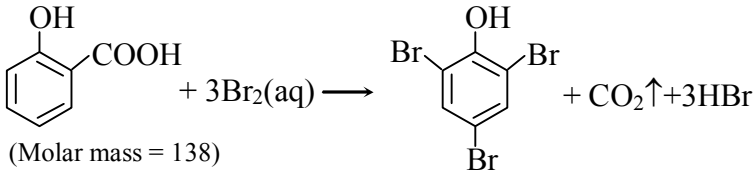
$$H_2A.xH_2O \xrightarrow{\Delta} H_2A + xH_2O$$

Mole of H_2O formed $= x \times$ mole of hydrated dibasic acid

$$\frac{(1.01 - 0.72)}{18} = x \times \frac{1.01}{143.14}$$

$x = 2.28$

degree of hydration is ≈ 2.

4. 

(Molar mass = 138)

Moles of salicylic acid $= \dfrac{276}{138} = 2$

Moles of Br_2 required = 3 × 2 = 6.

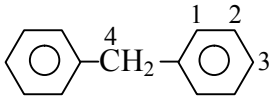
5. Let basicity of acid be n.

Equivalent of acid = Equivalent of KOH

$$\frac{1\times n}{146} = \frac{0.768}{56}$$

$$n = \frac{0.768\times 146}{56} = 2$$

6. Three in the ring and fourth on the CH_2 group as shown



Mathematics

PART – III

SECTION – A

1. Let x bullet are fired at A
E_1 – the target is at A, E_2 – the target at B
E – the target is hit

$P(E_1) = \frac{27}{28}$, $P(E_2) = \frac{1}{28}$, $P\left(\frac{E}{E_1}\right)$ = probability that target is hit when x bullet are fired at A

$= 1-\left(\frac{1}{3}\right)^x$

$P\left(\frac{E}{E_2}\right) = 1-\left(\frac{1}{3}\right)^{47-x}$

$P(E) = P(E_1)\cdot P\left(\frac{A}{E_1}\right) + P(E_2)\cdot P\left(\frac{A}{E_2}\right) = \frac{27}{28}\cdot\left(1-\left(\frac{1}{3}\right)^x\right) + \frac{1}{28}\left(1-\left(\frac{1}{3}\right)^{47-x}\right)$

$= 1-\frac{27}{28}\cdot 3^{-x} - \frac{1}{28}\cdot 3^{x-47}$

$\therefore \frac{d}{dx}(P(E)) = \frac{27}{28}3^{-x}\ln 3 - \frac{1}{28}3^{x-47}\ln 3$

$\frac{d^2}{dx^2}(P(E)) = -\frac{27}{28}(\ln 3)^2 3^{-x} - \frac{1}{28}(\ln 3)^2 3^{x-47} < 0$

For maximum P(E) $\Rightarrow \frac{d}{dx}(P(E)) = 0 \Rightarrow 27\cdot 3^{-x} = 3^{x-47} \Rightarrow x = 25$

2. For $0 \le r \le n$, the polynomial ${}^xC_r = \frac{x(x-1).....(x-r+1)}{r!}$ is of degree r
$\Rightarrow Q(x) = {}^xC_0 + {}^xC_1 + ... + {}^xC_n$ is polynomial of degree n
$Q(k) = 2^k$ for k = 0, 1, 2, ..., n
$\Rightarrow P(x) = Q(x)\ \forall\ x$ then
$\Rightarrow P(n+1) = Q(n+1) = 2^{n+1} - 1$

3. If a polynomial $a_0x^n + a_1x^{n-1} + ... + a_n$ is such a polynomial,
We can assume $a_0 = 1$, let $r_1, r_2, ..., r_n$ are its roots
$\Rightarrow r_1^2 + r_2^2 + ... + r_n^2 = a_1^2 - 2a_2$, $r_1^2r_2^2 ..... r_n^2 = a_n^2$

$\Rightarrow AM \ge GM \Rightarrow \frac{a_1^2 - 2a_2}{n} \ge (a_n)^{2/n}$

$\because a_1, a_2 = \pm 1$, we must have $a_2 = -1$, and $n \le 3$
$\Rightarrow$ Polynomials are
$\pm(x-1), \pm(x+1), \pm(x^2+x-1), \pm(x^2-x-1), \pm(x^3+x^2-x-1), \pm(x^3-x^2-x+1)$

4. For n = 1, 2, 3, 4, 5, 6, $\frac{f(n+1)}{f(n)}$ = $3, \frac{8}{3}, \frac{22}{8}, \frac{65}{22}, \frac{209}{65}, \frac{732}{209}$

The minimum of these is $\frac{8}{3}$, for n > 6, $\frac{f(n+1)}{f(n)} > 3 > \frac{8}{3}$

5. Let 1 + sin x = y

$I = \int \frac{1}{y}\sqrt{\frac{3-y}{1+y}}dy$ again let $\frac{3-y}{1+y} = t^2$

$I = 8\int \frac{t^2}{(t^2-3)(t^2+1)}dt = \sqrt{3}\ln\frac{t-\sqrt{3}}{t+\sqrt{3}} + 2\tan^{-1}t + c$

6. Let $a = \left(\sqrt{108}+10\right)^{1/3}$, $b = \left(\sqrt{108}-10\right)^{1/3}$

$a - b = x$, $a^3 - b^3 = 20$, $ab = 2$

$x^3 = (a-b)^3 = a^3 - b^3 - 3ab(a-b) = 20 - 6x$

$\Rightarrow x^3 + 6x - 20 = 0$

$\Rightarrow x = 2$ is only one real solution

7. $c_1c_2 = 1000 - 320 = 680$

$c_1p = \sqrt{(c_1c_2)^2 - (320)^2} = 600$, $PQ = r_1 - c_1p = 1000 - 600 = 400$

8. $S = \alpha + \beta = -1$

$P = \alpha\beta = z^3\left(\frac{z^{2n}-1}{z^2-1}\right)^2 = \frac{z}{(z+1)^2} = \frac{1}{z+\frac{1}{z}+2} = \frac{1}{4}\sec^2\left(\frac{\pi}{2n+1}\right)$

9. $f_6(f_m(x)) = f_4(x) = \frac{1}{1-x}$

$f_6(f_m(x)) = \frac{f_m(x)-1}{f_m(x)} = \frac{1}{1-x}$

Let $f_m(x) = k \Rightarrow f_m(x) = \frac{x-1}{x} = f_6(x) \Rightarrow m = 6$

Assign, $f_n(f_4(x)) = f_3(x) = \frac{1}{x}$

$\Rightarrow f_n\left(\frac{1}{1-x}\right) = \frac{1}{x} \Rightarrow$ let $\frac{1}{1-x} = t \Rightarrow x = \frac{t-1}{t}$ hence, $n = 5$

10. Given equation can be written as $\int_0^x xf(t)dt - \int_0^x t\,f(t)dt = \int_0^x f(t)dt + \sin x + \cos x - x - 1$ differentiating

$x\,f(x) + \int_0^x f(t)dt - x\,f(x) = f(x) + \cos x - \sin x - 1$ again differentiating

$f(x) = f'(x) - \sin x - \cos x$ which is linear differential equation of 1st order whose solution gives

$ye^{-x} = -e^{-x}\cos x + c$

$x = 0, y = 0 \Rightarrow c = 1$

$f(x) = e^x - \cos x$

11. Let $f(1) = t$

Put $x = 1$ in $f(x)f\left(f(x)+\frac{1}{x}\right) = 1$ ..... (1)

We get

$tf(t+1) = 1 \Rightarrow t \neq 0$ and $f(t+1) = \frac{1}{t}$

Put $x = t + 1$ in equation (1), we get $f\left(\frac{1}{t}+\frac{1}{t+1}\right) = t = f(1)$

$\Rightarrow \frac{1}{t}+\frac{1}{t+1}=1 \Rightarrow t=\frac{1\pm\sqrt{5}}{2}$

But for $t=\frac{1+\sqrt{5}}{2}$, $t > 1 – f(1) < f(1 + t) = \frac{1}{t} < 1$ a contradiction

12. Let M (line be $\bar{r}=\mu(ai+bj+ck)$ then acute angle between L and M is

$$\cos\alpha=\frac{(a+c)\cos\theta+\sqrt{2}b\sin\theta+(a-c)\sqrt{3}}{2\sqrt{2}\sqrt{a^2+b^2+c^2}}$$

$\alpha=\frac{\pi}{6}$, a + c = 0, b = 0, c = –a

$\bar{M}=\mu(i-k)$

13. M meets the given plane in $P_1, \left(2\sqrt{3}, 0, -2\sqrt{3}\right)$ line L meets the plane in

$P\left(2\cos\theta+2\sqrt{3}, 2\sqrt{2}\sin\theta, 2\cos\theta-2\sqrt{3}\right)$

Thus $(PP_1)^2 = 8$

$\Rightarrow$ P describes a circle of radius $2\sqrt{2}$

14. In radians we can say

$$\sin x=\sin\frac{\pi x}{180}=\sin\left(\pi-\frac{\pi x}{180}\right) \text{ or } \sin\left(2\pi+\frac{\pi x}{180}\right)$$

$$x=\frac{180\pi}{180+\pi} \text{ or } x=\frac{360\pi}{180-\pi}$$

On comparing with $\frac{p\pi}{q+\pi}$ and $\frac{m\pi}{n-\pi}$

We have, m = 360, n = 180, p = 180, q = 180

$\therefore$ m + n + p + q = 900

15. $\lim\limits_{x\to\infty}\frac{\left(A-C^2\right)x^2+Bx}{x\left[\sqrt{A+\frac{B}{x}}+C\right]}=2$ for existence of limit $A = C^2$

Hence, $\frac{BC}{A}=\frac{B}{C}$ (using $A = C^2$)

$\therefore L=\frac{B}{\sqrt{A}+C}=2$

If $C = -\sqrt{A}$ then limit does not exist hence $C=\sqrt{A}$

$\therefore \frac{B}{C}=4$

16. $g'(x) = –2f'(10 – 2x) \Rightarrow g'(L = 2) = –2f'(6) = 0$

SECTION – B

1. (A) $\left(\sqrt{x^2-5x+6}+\sqrt{x^2-5x+4}\right)^{x/2}=A$

$\left(\sqrt{x^2-5x+6}-\sqrt{x^2-5x+4}\right)^{x/2}=B$

Now, A + B = $2^{\frac{x+4}{4}} = 2\cdot 2^{\frac{x}{4}}$, $AB = 2^{\frac{x}{2}}$

$\Rightarrow$ A – B = 0 $\Rightarrow$ A = B

Since power of A and B are same

$\Rightarrow x = 0,\ \sqrt{x^2-5x+4} = 0 \Rightarrow x = 0, 4, 1$

(B) Equation can be written as $\left(\frac{x^2+x+2}{x^2+x+1}\right)^2 - (a-3)\left(\frac{x^2+x+2}{x^2+x+1}\right) + (a-4) = 0$

$t = \frac{x^2+x+2}{x^2+x+1}$ as $t \in \left(1, \frac{7}{3}\right]$

Equation reduces to $t^2 - (a-3)t + (a-4) = 0$ at least one root of equation lies in $\left(1, \frac{7}{3}\right]$

$t = a-4, 1$

$\Rightarrow$ One root lies in $\left(1, \frac{7}{3}\right] \Rightarrow 1 < a-4 \le \frac{7}{3} \Rightarrow a \in \left(5, \frac{19}{3}\right]$

(C) $(1+x)^2\left(1+\frac{x}{2}\right)^2 .....\left(1+\frac{x}{2^3}\right)^2 ..... = P_0 + P_1x + ... + P_rx^r .....$ ..... (1)

Replace x by $\frac{x}{2}$

$\left(1+\frac{x}{2}\right)^2\left(1+\frac{x}{2^2}\right)^2 .....\left(1+\frac{x}{2^n}\right)^2 = P_0 + P_1\left(\frac{x}{2}\right) + ... + P_rx^n + ...$

Equating the coefficient of x^r

$\Rightarrow P_r = \frac{2^2(P_{r-1}+P_{r-2})}{2^r-1}$ putting x_0 in (1), we get $P_0 = 1$

$\Rightarrow P_1 = 2 + 1 + \frac{1}{2} + \frac{1}{2^2} + ... = 4$

(D) x + y + z = n, $x \ge 1, y \ge 1, z \ge 1$ number of solution is $P_n = {}^{n-1}C_{n-3}$

$$\Delta = \begin{vmatrix} {}^{n-1}C_{n-3} & {}^{n}C_{n-2} & {}^{n+1}C_{n-1} \\ {}^{n}C_{n-2} & {}^{n+1}C_{n-1} & {}^{n+2}C_{n} \\ {}^{n+1}C_{n-1} & {}^{n+2}C_{n} & {}^{n+3}C_{n+1} \end{vmatrix} = \frac{1}{8}\begin{vmatrix} (n-1)(n-2) & n(n-1) & n(n+1) \\ n(n-1) & n(n+1) & (n+1)(n+2) \\ n(n+1) & (n+2)(n+1) & (n+3)(n+2) \end{vmatrix}$$

On applying $C_3 \to C_3 - C_2$, $C_1 \to C_1 - C_2$ and $C_1 \to C_1 + C_3$ = $\frac{1}{2}\begin{vmatrix} 1 & n(n-1) & n \\ 0 & 2n & 1 \\ 0 & 2(n+1) & 1 \end{vmatrix}$ = –1

$\therefore |\Delta| = 1$

2. (A) $z = (2\cos 30^\circ \cdot \cos 18^\circ + i\cdot 2\sin 30^\circ \cdot \cos 18^\circ)^6$
$= 2^6 \cos^6 18^\circ (\cos 180^\circ + i \sin 180^\circ) = -2^6 \cos^6 18^\circ$

(B) $f'(x) + f^2(x) = -1 \Rightarrow \frac{dy}{dx} + y^2 = -1 \Rightarrow y = \tan(c-x)$

$\therefore$ We must have $c - a = (2n+1)\frac{\pi}{2}$ and $c - b = (2n-1)\frac{\pi}{2}$ for some $n \in I$

$\therefore \min\left(\frac{b-a}{\pi}\right) = 1$

(C) Let $\cos^{-1} x = \theta$ then $\tan\left(\frac{\pi}{4}+\frac{\theta}{2}\right) + \tan\left(\frac{\pi}{4}-\frac{\theta}{2}\right) = 1$

$\Rightarrow 2\sec\theta = 1$

$\Rightarrow \sec = \frac{1}{2}$ no solution

(D) $\theta = \frac{\pi}{7} \Rightarrow 7\theta = \pi \Rightarrow 4\theta = \pi - 3\theta \Rightarrow \tan 4\theta = -\tan 3\theta$

$\Rightarrow t^3 - 21t^2 + 35t - 7 = 0$, $t = \tan^2\theta$

Above equation has the roots $\tan^2\frac{\pi}{7}$, $\tan^2\frac{2\pi}{7}$, $\tan^2\frac{3\pi}{7}$ and sum of roots = 21

Replacing $t = \frac{1}{r}$

$7r^3 - 35r + 21r - 1 = 0$ has three roots

$\cot^2\frac{\pi}{7}, \cot^2\frac{2\pi}{7}, \cot^2\frac{3\pi}{7}$ and sum of roots = 5

SECTION – C

1. Let $g(x) = \int \frac{f(x)dx}{x^2(x+1)^3}$ ..... (1)

$= \int \left(\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1} + \frac{D}{(x+1)^2} + \frac{E}{(x+1)^3} \right) dx$

$= A\ln x - \frac{B}{x} + C\ln(1+x) - \frac{D}{1+x} - \frac{E}{2(x+1)^2}$

Since g(x) is a rational function hence A = C = 0

$g(x) = \int \left(\frac{B}{x^2} + \frac{D}{(x+1)^2} + \frac{E}{(x+1)^3} \right) dx$ ..... (2)

Comparing (1) and (2), we get $f(x) = (B + D)x^3 + (3B + D + E)x^2 + 3Bx + B$

$\Rightarrow$ f(x) is quadratic

$\Rightarrow B + D = 0$, $f(0) = 1$ gives $B = 1 \Rightarrow D = -1$

$\therefore f(x) = (2 + E)x^2 + 3x + 1$

$f'(0) = 3$

2. $\begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ [x] & [y] & [z]+1 \end{vmatrix}$ $R_1 \to R_1 - R_3, R_2 \to R_2 - R_3$ on solving [x] + [y] + [z] + 1, taking maximum value, we get 4

3. $\alpha + \beta + \gamma = \tan^{-1}\left(\frac{36}{77}\right) + \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{8}{15}\right) = \frac{\pi}{2}$

$\Rightarrow \sum \cot\alpha = \prod \cot\alpha$ and hence $A = \frac{\sum \cot\alpha}{\prod \cot\alpha} = 1$

and $\sum \tan\alpha\tan\beta = 1 \Rightarrow A + B = 2$

4. A = f(–1) = 19

B = g(–9) = 2 ln 3

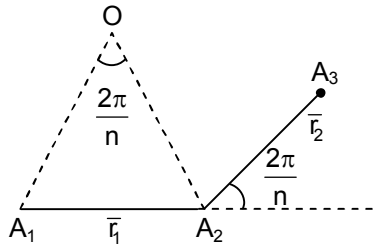
$C = \int_{-2}^{1} \frac{3(x^2 + 2x - 2)}{x^3 + 3x^2 - 6x + 11} dx = \ln 9 - \ln 27 = -\ln 3$

$D = \frac{d}{dt}\int_1^{t^2} g(x)\,dx = 6g(9) = 12\ln 3$

$A(B + C) - D = 7\ln 3 \Rightarrow k = 7$

5. $|r_1 \times r_2| = |r_1||r_2|\sin\frac{2\pi}{n}$

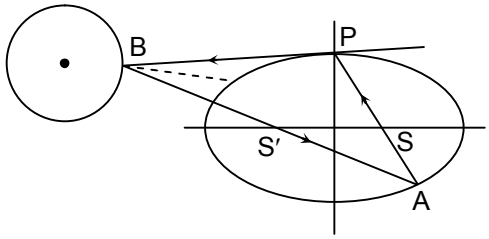
$|r_1 \cdot r_2| = |r_1||r_2|\cos\frac{2\pi}{n}$



The given equality reduces to $\tan\frac{2\pi}{n} = 1$

$\Rightarrow \frac{2\pi}{n} = m\pi + \frac{\pi}{4} \Rightarrow n = \frac{8}{1+4m}\,(m \in z)$, least n = 8

6. Using reflection property of ellipse equation of reflected ray SP is



$4x + 3y = 12$ ….. (1)

Solving equation (1) with ellipse, we get

$A \equiv \left(\frac{75}{17}, -\frac{32}{17}\right)$

Equation of reflected ray BS′ is

$16x + 63 + 48 = 0$ ….. (2)

Equation of incident ray PB is

$y = 4$ ….. (3)

Equation of bisector of (2) and (3) with negative slope gives normal to circle at 3 which is

$4x + 32y - 53 = 0$

Point B is $\left(-\frac{75}{4}, 4\right)$, using parametric equation of line ordinate of centre is 5